Aaradhya Dev Tamrakar

AI/ML Engineering · AI Workflow Automation · Embedded AI · Model Evaluation

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Final-year Electronics, Communication and Information Engineering student at IOE, Kathmandu Engineering College, building applied ML systems, AI workflow tools, and embedded ML software. Fuse AI Fellow (Fusemachines) and NSSR DataCamp Fellow. I use AI to accelerate development, then turn recurring work into reusable tools and deterministic checks — reducing the need to solve the same task in another prompt. Experienced across demographic fairness auditing, multi-provider coordination, edge ML inference, and developer automation.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Co-developed **BiasAperture**, an automated demographic fairness auditing pipeline measuring predictive disparities on computer vision benchmarks as the fellowship capstone project.
- Built applied agentic and ML coursework pipelines including a 5-stage Text-to-SQL system (Planner → Generator → Validator → Executor → Summarizer) tested against structured evaluation suites.
- Developed customer churn and lifetime value (CLV) prediction workflows utilizing **SMOTE**, cost-sensitive threshold tuning, and **SHAP** explainability.
- Implemented MLOps coursework for experiment tracking, containerized microservices (**FastAPI**, **Docker**, **PostgreSQL**), and scheduled data-drift evaluation.
- Evaluated probabilistic estimation and time-series forecasting ensembles on retail datasets as part of advanced predictive modeling modules.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed applied GenAI workflows automating data cleaning, schema normalization, and exploratory analysis.
- Completing advanced technical pathways in SQL database design, Python data analysis, and predictive modeling.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026

Python · AIF360 · Fairlearn · Statistical Testing · Jinja2 · OpenCV

- Co-developed an automated prediction-auditing and reporting pipeline as a two-person fellowship capstone with Tisha Manandhar.
- Audited classification performance across **126 demographic bins** on the FairFace benchmark (**97,698 images**), computing Disparate Impact Ratio, Demographic Parity Difference, and Equalized Odds.
- Engineered self-contained HTML and PDF report generation with confidence intervals and diagnostic disparity plots; matched model comparison and validation remain active work.

SPARK — Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TensorFlow Lite Micro · Python · FreeRTOS · MPU-6050

- Project lead for ML modeling, gateway processing, and explainability in a four-person team; hardware and firmware have dedicated owners.
- Trained 1D-CNN fall-detection models on the SisFall benchmark (38 subjects, 38,420 windows), evaluating subject-stratified cross-validation with **0.9185 AUC-ROC** and 87.81% sensitivity.
- Quantized model to an **18.5 KB INT8 CNN** footprint for ESP32-S3 edge deployment; on-device hardware testing, BLE streaming, and volunteer cohort validation remain pending.

Fleet Orchestrator — AI Workflow Coordination & Tooling | 2025 – Present

Python · FastAPI · Provider Adapters · Model Context Protocol (MCP) · SQLite

- Built coordination tooling for multi-step AI workflows, with task state, worker adapters, scheduling, and live status inspection.
- Standardized provider adapters (local models, CLI harnesses, API endpoints) with reusable interfaces to eliminate repeated orchestration scripts.
- Integrated local MCP tool servers and automated execution logging to ensure reproducible batch processing and reliable error recovery.

Agent-Customization-Sync — Cross-IDE Configuration Tooling | 2025 – Present

Python · SHA-256 · CLI · Configuration Governance

- Built configuration snapshots, guarded restore, rules transpilation, and SHA-256 drift audits across development environments (Antigravity, Cursor, Claude Code).
- Audited instruction overhead and linked shared skills to reduce configuration duplication and maintain environment consistency across machines.

Gesture-Controlled Self-Balancing Robot (GCSBR) | 2024 – 2025

Arduino · C++ · MediaPipe Hands · CameraX · Kotlin · PID Control

- Implemented real-time PID stabilization firmware on an inverted pendulum robot, tuning balance control parameters for stable upright recovery.
- Developed an Android companion app (Kotlin, MediaPipe Hands, CameraX) mapping hand gestures to directional actuation commands over Bluetooth.

Super-NLM — Multi-Account NotebookLM Aggregator & MCP Tool | 2025 – Present

Python · Model Context Protocol (MCP) · Google Drive API · FastAPI

- Developed an MCP tool server enabling AI agents to query notebooks, synchronize Google Drive research folders, and download Studio artifacts.
- Implemented queue scheduling and quota-aware backoff handling to support automated document synthesis across study materials.

Win-Vault — Native Windows Folder Privacy Utility | 2025

Python · Win32 APIs · NTFS ACLs · PowerShell

- Engineered a zero-dependency Windows folder locking utility enforcing access restrictions via native NTFS Access Control Lists (ACLs).
- Eliminated third-party driver overhead by utilizing standard operating system security permissions and PowerShell automation.

NovaOptimizer / System-Optimizer — Workstation Memory Utility | 2025

Python · Win32 APIs · PowerShell · CLI

- Built a native Windows performance utility reclaiming **1.2–3.4 GB working-set RAM** under heavy multitasking developer workloads.
- Automated background cleanup routines via standard Windows process memory APIs without disrupting active IDE sessions.

nepali-ocr-ai — Low-Resource Document Processing Pipeline | 2025

Python · Tesseract OCR · OpenCV · Devanagari NLP

- Prototyped a document intelligence pipeline combining OCR with Devanagari text normalization and Preeti ↔ Unicode font transcoding.
- Built preprocessing filters in OpenCV for uneven illumination and historical print noise reduction.

Utility MCP Suite — LocalSend & Developer Bridges | 2025 – Present

Python · Model Context Protocol (MCP) · FastAPI · Networking

- Developed custom MCP servers including a LocalSend bridge enabling AI agents to discover LAN peers and transfer build artifacts securely.
- Built headless IPC connectors enabling agent-assisted operations with local design and engineering applications.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Coordinate branch planning, technical workshops, and cross-committee communication for Kathmandu Engineering College IEEE student body.
- Facilitated technical awareness events and student engagement initiatives across engineering departments.

Makerspace Ambassador — KEC Maker's Space, Kathmandu Engineering College | June 2026 – Present

- Maintain website content, interface fixes, and documentation for student prototyping equipment.
- Support students with 3D printing, embedded electronics prototyping, and lab equipment access.

Event Manager (2026 – Present); Resource Manager (2024 – 2026) — Electronics Project Club (EPC), Kathmandu Engineering College | 2024 – Present

- Manage event logistics, hardware inventory, and project exhibitions for the college electronics department.
- Organized project showcases and hardware component distribution for student engineering competitions.

TECHNICAL SKILLS

Machine Learning & Evaluation: Python, PyTorch, scikit-learn, Pandas, NumPy, XGBoost, SHAP, Fairlearn, AIF360, OpenCV, MediaPipe

AI Systems, Backend & Tooling: FastAPI, Docker, Model Context Protocol (MCP), SQL (PostgreSQL, SQLite), asyncio, REST APIs, MLflow

Embedded Systems & Edge AI: C/C++, ESP32-S3, TensorFlow Lite Micro, FreeRTOS, Arduino, Sensor Interfacing (IMU, UART/I2C/SPI), Kotlin

Engineering Practice & Tooling: Git, GitHub, Linux, PowerShell, Pytest, Reproducible Configurations, WordCraft, LaTeX

CERTIFICATIONS, WORKSHOPS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) • IEEE SPAx (Student Professional Awareness Venture) Organizing Committee (2025 – 2026) • IEEEXtreme 19.0 Programming Competition (2025) • PCB Design & Fabrication Workshop (2025) • AWS Cloud Computing Workshop (2024) • Prompt Engineering Fundamentals (2024) • Microsoft Learn Student Ambassador (2024 – 2025)

ACADEMIC & PROFESSIONAL REFERENCES

Er. Suramya Sharma Dahal — Head of Department, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University) • Email: surammya@gmail.com. Context: Academic department head overseeing undergraduate engineering curriculum and senior projects

Er. Dipen Manandhar — Faculty Member & Major Project Supervisor, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University). Context: Undergraduate major capstone project supervisor (SPARK edge-AI wearable)

Shreejan Kisee — Teaching Assistant & Capstone Supervisor

Fusemachines AI Fellowship. Context: AI Fellowship capstone supervisor (BiasAperture demographic auditing) and coursework mentor